Module 10: Linux Exploitation and Privilege Escalation

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# Exercise 06: Using GTFO to Search Binaries
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Scenario

Sudo Rights Enumeration & Binary Shell Breakout (`sudo pkexec`) exploits administrative permissions delegated to unprivileged users via the `/etc/sudoers` file. When an account possesses `sudo` execution rights for utilities listed in GTFOBins (such as `pkexec`, `env`, `find`, `vim`, or `awk`), standard shell restriction boundaries can be bypassed instantly to obtain full root subshells.

Key technical concepts in sudo enumeration & binary exploitation:
1. **Sudo Privilege Enumeration (`sudo -l`)**: Executing **sudo -l** to query the security policy for the current user (**ubuntu**), revealing permitted commands, target binaries, and execution rules.
2. `pkexec` **Sudo Shell Breakout**: Executing **sudo pkexec /bin/sh** to leverage root privileges via PolicyKit's execution wrapper, spawning an elevated shell.
3. **Interactive Root Subshell Initialization**: Upgrading the spawned shell to an interactive tty instance using **bash -i**.
4. **Privilege & System Verification**: Auditing effective user identity (**whoami** $\rightarrow$ `root`) and reviewing root file system contents (**ls -la /root**).

In this exercise, we log into **Ubuntu** VM (**ubuntu** / **toor**), execute **sudo pkexec /bin/sh**, spawn interactive subshell **bash -i**, verify root privileges (**whoami**), inspect directory contents (**ls**), open a new terminal, and enumerate permitted sudo commands using **sudo -l**.

Objective

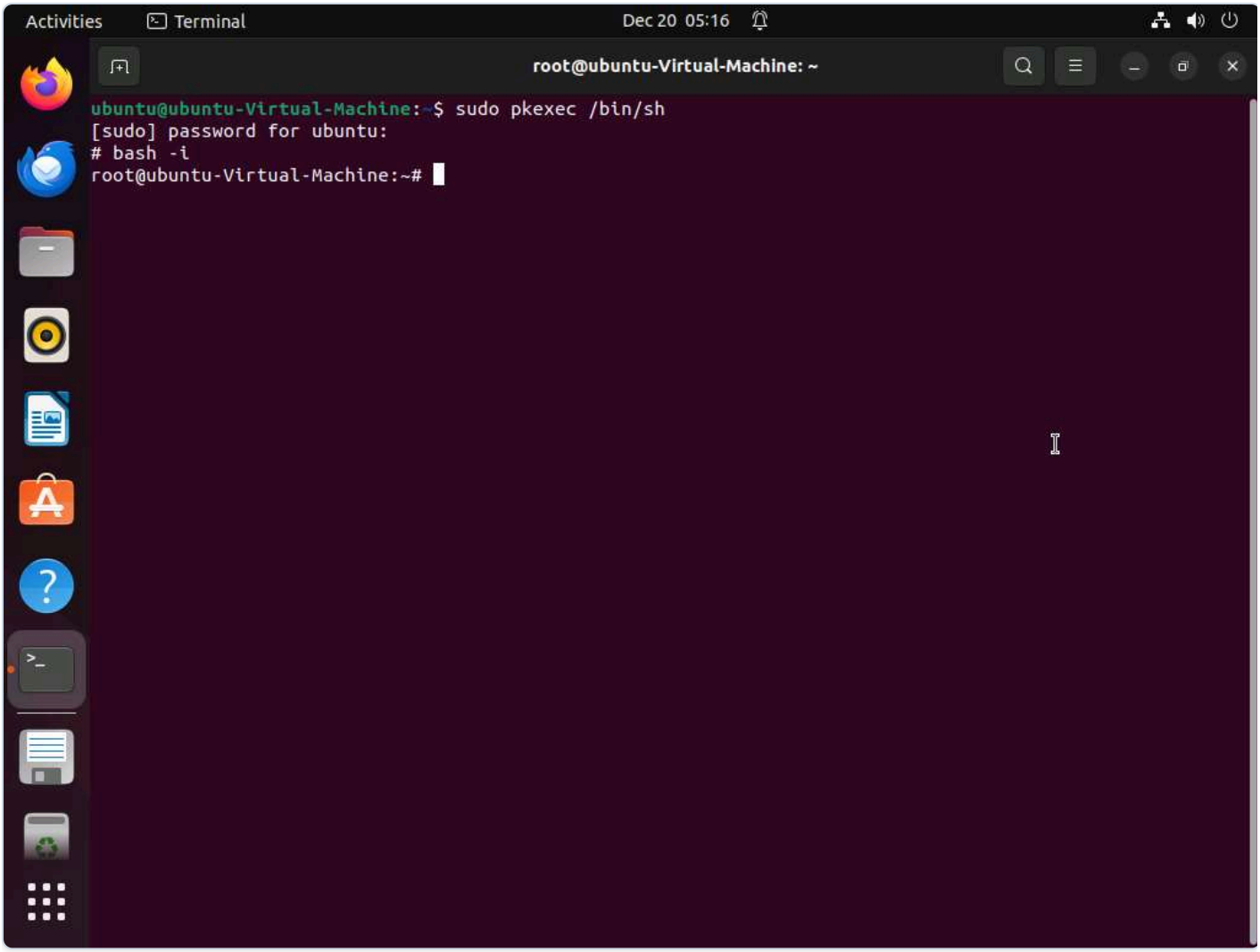
The objective of this lab is to help students learn how to:
- Authenticate to Ubuntu VM (**ubuntu** / **toor**)
- Execute sudo binary shell breakout via pkexec: **sudo pkexec /bin/sh**
- Initialize interactive Bash subshell: **bash -i**
- Verify active root user identity: **whoami**
- List root directory contents: **ls**
- Enumerate permitted sudo execution rights for current user: **sudo -l**

Task 1: Leverage Sudo Pkexec for Root Shell Breakout and Enumerate Sudo Rights with Sudo -l

In this task, we execute `sudo pkexec /bin/sh`, initialize interactive shell `bash -i`, verify root access, list root contents, and inspect permitted sudo commands via `sudo -l`.

Lab Tasks

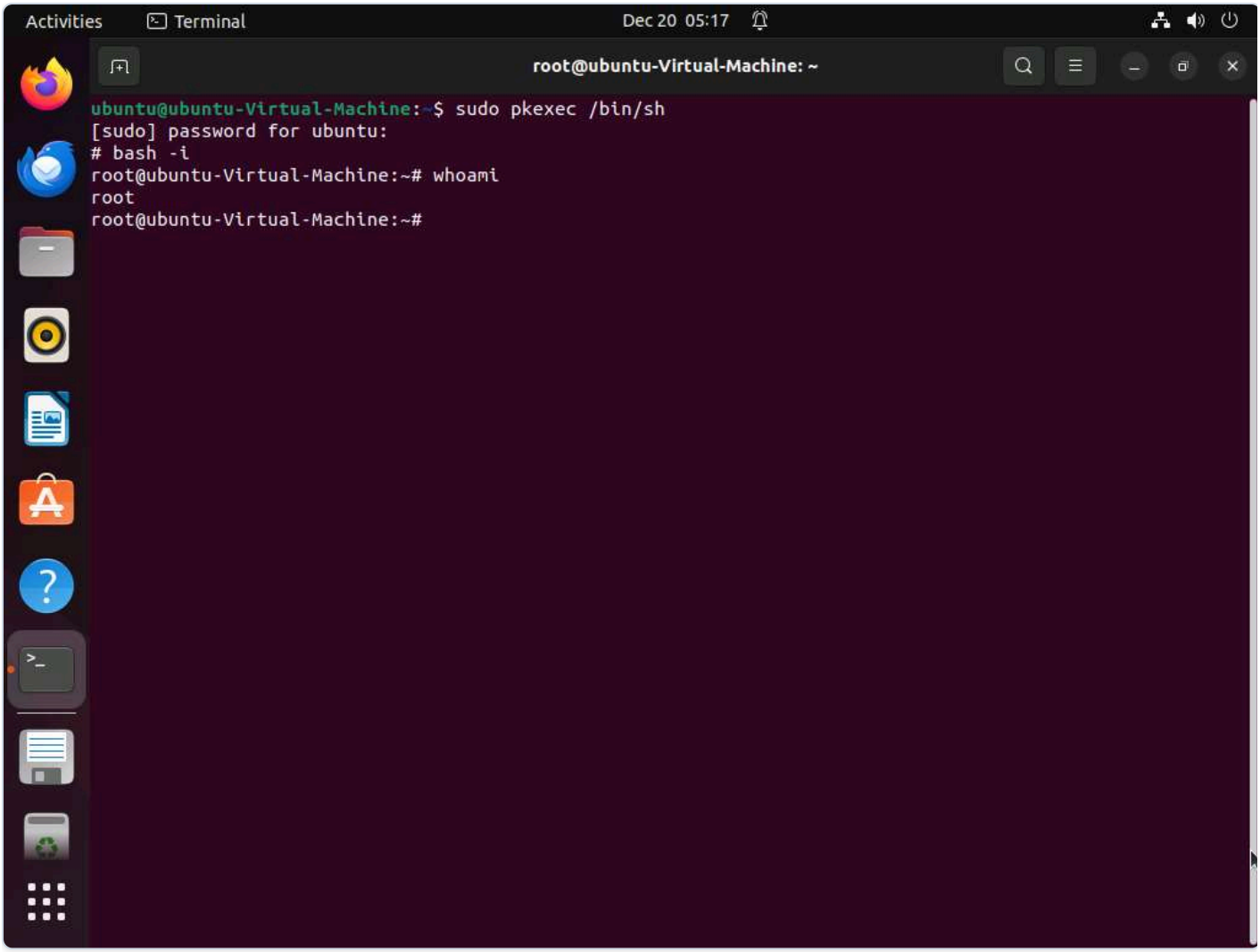
1. Log into **Ubuntu** VM using credentials (**ubuntu** / **toor**). Open terminal window and attempt sudo shell breakout via PolicyKit wrapper:
   sudo pkexec /bin/sh
   When prompted for password, enter **toor**. Inside the shell, launch an interactive Bash subshell:
   bash -i



1. Verify effective user identity to confirm root privilege escalation:

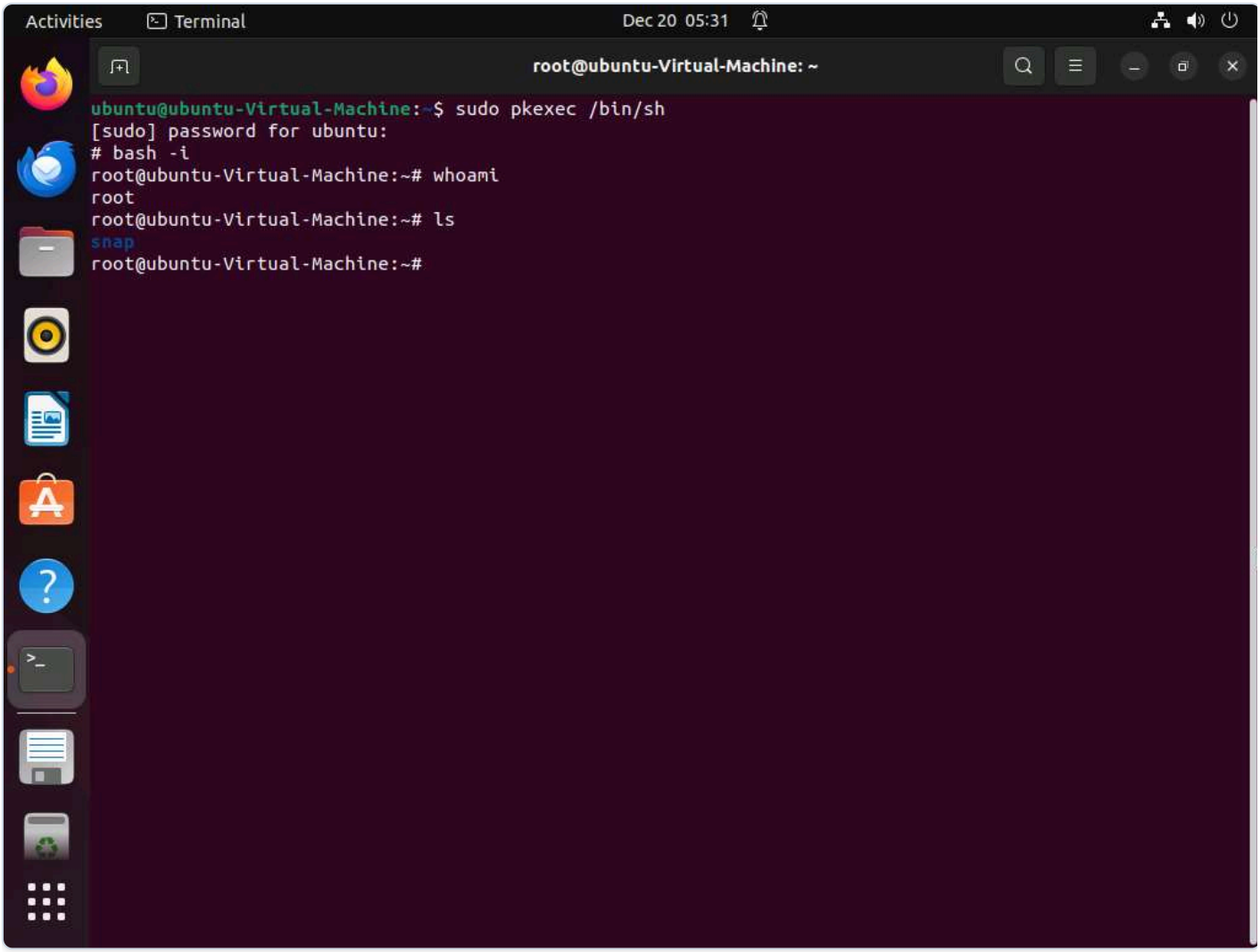
   whoami

   Observe output: `root` user identity maintained.

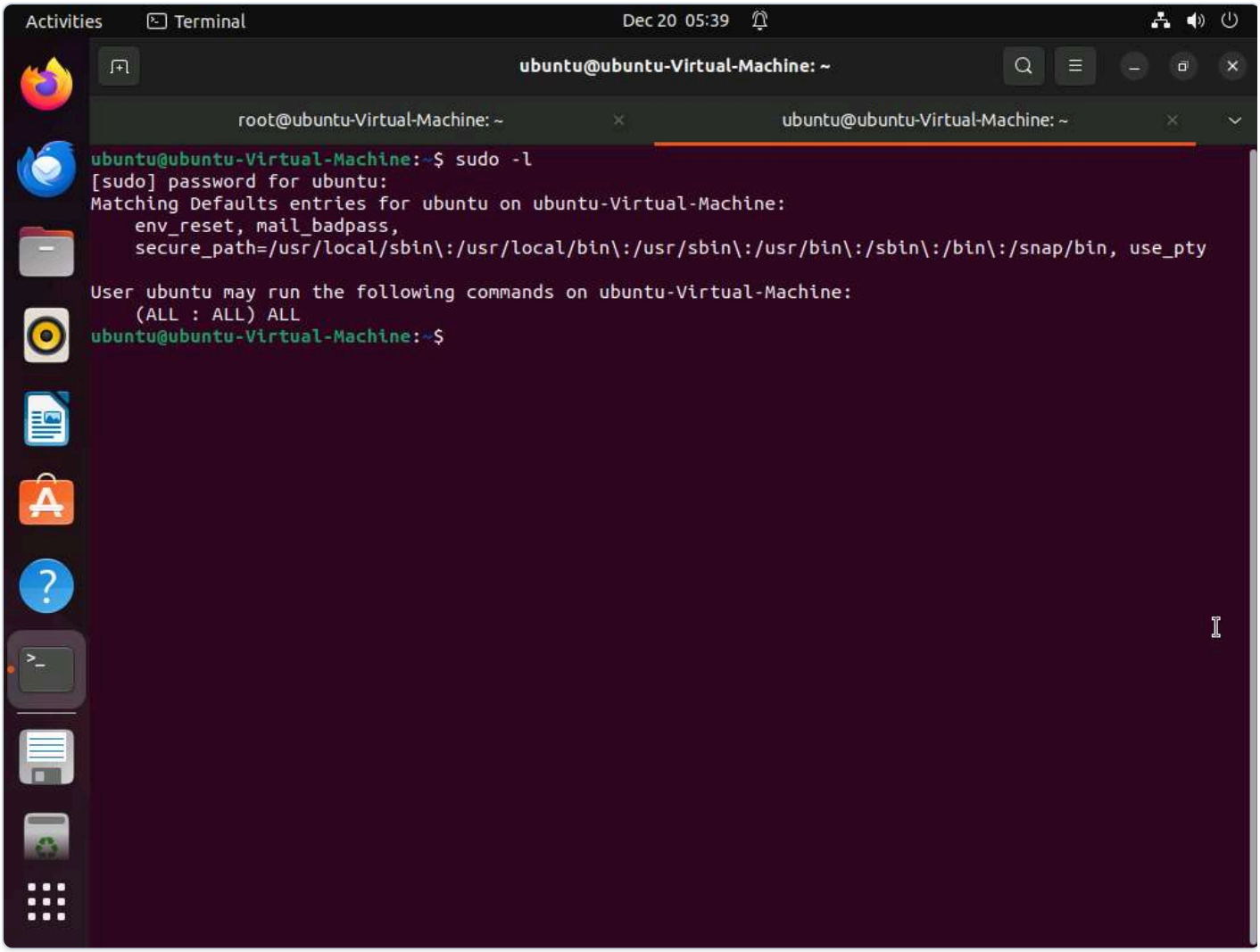


1. List contents of the root filesystem directory:

   ls



1. Open a new terminal window and enumerate all commands allowed for current user under sudo security policy:
   sudo -l
   Enter password **toor** when prompted.



This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 10.6.1.1** On an Ubuntu machine, use GTFO to identify binaries that can be easily exploited, allowing direct access to a root shell with a single command. Enter the command is used to list the sudo privileges available to the current user. [!SUCCESS] ✅ **Answer:** **sudo -l**